

University of Barishal
Department of Computer Science and Engineering
Course Title: Object Oriented Programming
Course Code: CSE-2107
2nd Year 1st Semester Final Examination
Admission Session: 2018-19

Asif Sir

Time: 3 Hours

Marks: 60

N.B.: Answer any **FIVE** questions out of the followings. All parts of each question must be answered consecutively. Right side of the question shows the maximum marks.

1.a) Distinguish between the following terms and provide suitable examples for each: 8

- | | |
|-----------------------------------|---|
| i) Objects and Classes | ii) Data abstraction and data encapsulation |
| iii) Inheritance and polymorphism | iv) Dynamic binding and message passing |

b) What will be output of the below code? Write answer with explanation. 4

i) <pre>public class A { public static void main (String[] args) { if (5) break; } }</pre>	ii) <pre>public class A { public static void main (String[] args) { System.out.println('j' + 'a' + 'v' + 'a'); } }</pre>
iii) <pre>public class A { public static void main (String[] args) { int \$_ = 5; } }</pre>	iv) <pre>public class Demo{ public static void main (String[] arr){ } public static void main (String arr){ } }</pre>

2.a) What is a static variable? Explain the static method with a proper example. What is the main method is ~~not~~ public static in Java? 4

b) Find and fix errors in the following code segment. 4

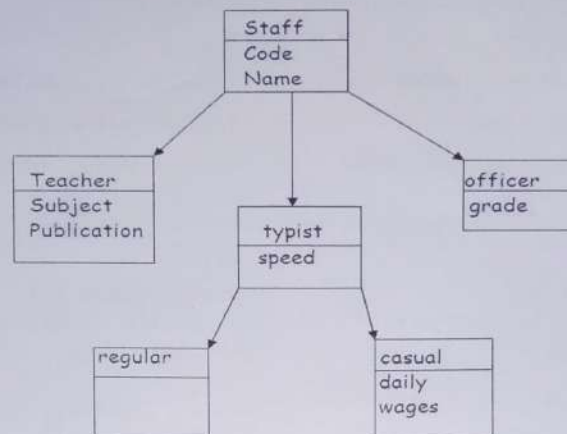
<pre>public abstract class Bike { private string Model {get; set;} protected abstract void Display(); }</pre>	<pre>public sealed class Hond { public int Speed {get;set;} = 160 protected override void Display() { Console.WriteLine(Model); } }</pre>	<pre>class Program { public static void Main (){ Bike bike = new Bike(); Bike bike = new Hond(); bike.Display(); Console.WriteLine(bike.S peed); } }</pre>
<pre>Public class Honda4: Hond { }</pre>		

c) Write down the output of the following programs: 4

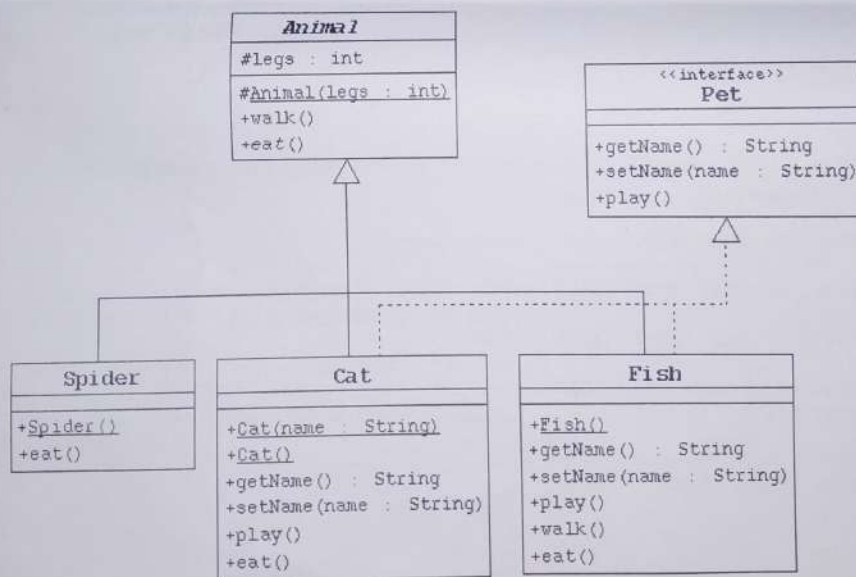
i) <pre>int main() { int i=10, j=5, modresult=0, divresult=0; modresult = i % j; cout<< modresult << " "; divresult= i / modresult; cout<< endl<< divresult; return 0; }</pre>	ii) <pre>int main() { int i[5] = { 5, 6, 7, 8, 5}, j; (i [1] == i[0]) ? (j=2) (j=1); if (j >= 1) cout<< "OK"; else cout<< "YES"; return 0; }</pre>
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- 3.a) What is multilevel inheritance? A class named **SuperClass** has a method **Display()** [It prints the message "I am super class"]. Another class named **ChildClass** inherits **SuperClass**. Now, **ChildClass** should have the same **Display()** method but it should print the message "I am child class". You need to implement this scenario using OOP. You can choose your favorite language for implementation.
- b) Barishal University expects to keep a database of its staffs. The database is separated into several classes whose hierarchical connections are shown in following figure. The figure also shows the minimum details required for each class. Specify all classes and define functions to create the database and retrieve individual information as and when required.



- 4.a) Explain the Is-a Relationships and Has-a Relationships in OOP. Give real life example of both relationships.
- b) In the following Class Diagram, you will see a hierarchy of animals that is rooted in an abstract class **Animal**. Several of the animal classes will implement an interface called **Pet**.



Create the **Animal** class, which is the abstract superclass of all animals.

- Declare a protected integer attribute called **legs**, which records the number of legs for this animal.
 - Define a protected constructor that initializes the **legs** attribute.
 - Declare an abstract method **eat**.
 - Declare a concrete method **walk** that prints out something about how the animals walks (include the number of legs).
- 5.a) How does **String** class differ from the **StringBuffer** class? What are the applications of wrapper classes?

The annual examination results of 50 students are tabulated as follows:

6

Roll No.	Subject-1	Subject-2	Subject-3

Write a program to read the data and determine the following:

- Total marks obtained by each student.
- The highest marks in each subject and the Roll No. of the student who secured it.
- The student who obtained the highest total marks.

c) Correct the code to eliminate the compile time error thrown.

2

```
class Palindrome {
    public static void main(String[] arr){

        String palindrome = new String ("Rod saw I was Dor");
        int len = palindrome.length();
        String dest = new String();
        for(int i = (len - 1); i >= 0; i--) {
            dest.append(palindrome.charAt(i));
        }
        System.out.format("%s\n", dest.toString());
    }
}
```

a) Describe the various form of implementing interfaces. Give examples of Java code for each case.

4

b) In this code, the interface acts as a type. Will this code compile successfully? If not, correct the code.

2

```
interface NewShape
{
    void draw();
}

class NewCircle1 implements NewShape
{
    public void draw ()
    {
        System.out.println("New Circle-1 Drawn");
    }
}

class NewCircle2 implements NewShape
{
    public void draw ()
    {
        System.out.println("New Circle-2 Drawn");
    }
}

class InterfaceTest
{
    public static void main(String args[])
    {
        NewShape nc1 = new NewCircle1 ();
        NewShape nc2 = new NewCircle2 ();
        nc1.draw();
        nc2.draw();
    }
}
```

c) What is a Java Package and How is it used? Give example.

4

d) The method draw() in the NewCircle has been set as private. Class SamePackage is in the same package as NewCircle. Will the class be able to use the method?

2

```
package circle;
import circle.NewCircle;

class SamePackage
{
```

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```

public static void main(String args[])
{
    NewCircle nc = new NewCircle ();
    nc.draw();
}
}

```

- 5
- 2
- 7(a) Exemplify the complete life cycle of a thread.
- b) The code given below calls the run() method of two threads while setting their priority. Will this code compile successfully? If not, correct the code.

<pre> class t1 extends Thread { public void run () { System.out.println("Thread1 class"); } } </pre>	<pre> class t2 extends Thread { public void run () { System.out.println("Thread2 class"); } } </pre>
<pre> class ThreadTest { public static void main (String args []) { t1 t11 = new t1(); t2 t22 = new t2(); t11.setPriority(Thread.MIN_PRIORITY); t22().setPriority(Thread.MIN_PRIORITY); t11.run(); t22.run(); } } </pre>	

- c) Find errors in the following code which writes data of one file to another file. 2

```

import java.io.*;
public public static void main(String args []) {

    try {
        FileReader fr = new FileReader("in.dat" );
        FileWriter fw = new FileWriter("out.dat");
        int ch;
        while ((ch=fr.read()) != 1) {
            fr.write(ch);
        }
    } catch (Exception ex) {
        System.out.println(ex);
    }
}

```

- d) Create a try block that is likely to generate three types of exception and then incorporate necessary catch blocks to catch and handle them appropriately. 3
- 8(a) The Collections class of Java Collection Framework has several methods (e.g. sort(), min(), max()) that require natural ordering of the classes to be handled. Explain the two possible conditions that must be satisfied on the classes to be able to use the sort() method to arrange the list of objects these classes. 5
- b) Explain why you would choose a LinkedHashSet (or LinkedHashMap) over a HashSet (or HashMap, respectively). 3
- c) Distinguish between 4
- InputStream and Reader classes
 - OutputStream and Writer classes

"Every student can learn, just not on the same day, or the same way."
- George Evans